

Reach for the stars!

Long before the first spacecraft blasted off, people dreamed of travelling to other star systems in the hope of finding new planets. Travel between the stars is called interstellar travel, and one day it may be possible. But the difficulties are huge.

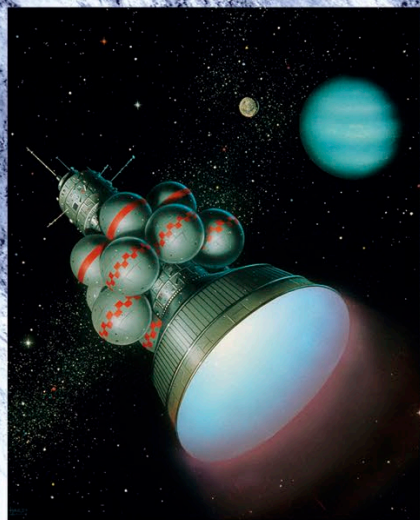
The main problem

Our fastest spaceships travel at significantly less than one thousandth of the speed of light. In terms of the vastness of space, that's incredibly slow. If we are to explore further, we need faster spacecrafts.

Space travel

A Russian physicist named Konstantin Tsiolkovsky had an idea for a spacecraft long before people went into space.

1903 model of
Tsiolkovsky's
futuristic spacecraft



Project *Daedalus* was planned as an unmanned space probe. It would travel at 12 per cent of the speed of light, reaching Barnard's Star within 50 years.

weird or what?

Daedalus wouldn't be able to stop at Barnard's Star, because slowing down would need as much fuel as speeding up. So, it would just whizz by taking pictures and measurements.

What's been done?

A long-term study into a possible interstellar spaceship took place in the 1970s. It was based on a mission to reach Barnard's Star, almost six light years away. The proposed spacecraft was named *Daedalus*.